

CELLPHONE SIGNAL BOOSTER

For Your Office

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This guide should help you select the most appropriate signal booster for your office

Mobile signal boosters are often used to overcome network issues inside office premises. Network unavailability might be caused due to the infrastructure or the location. With so many options of network boosters available in the market, it is difficult to make a choice. This article enlists the main parameters to help you choose the best signal booster for your office.

How signal boosters work

Cellphone signal boosters improve the weak indoor tower reception on your mobile device by amplifying signals received from outside the building. A signal booster setup primarily consists of an external antenna, one or more inside antennae, and an amplifier—all connected with the help of cables.

The external antenna is positioned on a building's terrace or outside a wall, while the other antenna is to be attached to the amplifier device indoors. The external antenna detects the network signal and passes it on to the amplifier through cables, where it is boosted. The amplified signal is forwarded to the internal antenna, which broadcasts it indoors. The signal flow route is reversed when you send some communication from your phone.

Broadly, signal boosters can be divided into two categories—one for building and office use, and the other for fixing on vehicles. In this article, we focus on antennae for small- and medium-sized office buildings.

Fig. 1: A network signal booster set with external and internal antennas, amplifier and cables



What features should you look for?

When selecting a signal booster, the three important factors to consider are the frequency and the network supported (2G, 3G or 4G and the network provider), amplifier strength and antennae.

Check network compatibility. Signal boosters support networks based on their frequencies. Network service providers tap into respective frequencies like 900MHz, 1800MHz, 2100MHz and so on to provide signals. Booster devices support these networks and their bands (2G, 3G or 4G) if they respond to the respective frequencies.

While a group of boosters in the market work with specific network providers, most of the devices support networks of almost all major carriers (Vodafone, Airtel, TATA and so on). Today, majority of boosters supporting dual-band frequencies (up to 2100MHz) are compatible with 2G and 3G. However, the selected booster should preferably support 4G, which calls for a higher-band device. Getting the highest frequency of network coverage definitely provides the biggest advantage, but the cost goes up.

Go for a strong amplifier. The amplifier, as clear from the name, amplifies the signal received from the antenna. The preferable strength of the amplifier can be determined by two factors: Desired coverage area in the building and the strength of the original signal outside. Amplification is measured in decibels (dB), which denotes relative strengthening of the signal.

For a medium-size office space of around 280-370sq.m (3000-4000sq.ft.), 60-65dB amplifier gain is a good choice. To get stronger amplification, go for amplifiers with higher decibel counts. Vendors also use another unit called decibel milliwatts (dBm) for signal output, which implies the actual amount of signal power available as output. The value of this parameter should be 20-23dBm.

Type of antennae. As mentioned earlier, the setup uses two types of antennae—one to be installed outside and the other to be used indoors. For office building applications, the key deciding factors for antennae include signal